Unit 1

1. Introduction to AI

Module 1.8

1.8 Pre-Trained Models

Tutorial 1.8.1

1.8.1 Introduction to Pre-Trained Models

AI-1.8.1 K2 Explain the use of pre-trained AI models and the risks associated with them.

Pre-Trained Models

- Machine learning models which are already trained on large datasets to learn general patterns and features.
- These models can be used by developers to further fine-tune for specific tasks which will save time, computing resources and effort to gather large data set for training model from scratch.
- The pre-Trained models are cheaper alternative for creating a new model that extends and/or focuses the functionality of the pre-trained model.
- Some Pre-Trained models in Computer Vision

Model Name	**Developer**	**Key Capabilities & Highlights**
YOLOv12	Ultralytics	Real-time object detection
SAM 2 (Segment Anything Model)	Meta	Prompt-based segmentation across domains
DINOv2	Meta	Self-supervised vision encoder
MaxViT	Google	Combines grid and window attention for efficient spatial reasoning

Pre-Trained Models

Few Pre-Trained Natural Language Processing (NLP) Models

Model Name	Developer	Key Capabilities & Highlights
Gemini Ultra	Google DeepMind	Multimodal LLM with advanced reasoning, multilingual fluency, and real-time search integration
GPT-4 Turbo	OpenAI	Faster, cheaper variant of GPT-4 with extended context window and improved instruction following
Mistral 7B	Mistral AI	Lightweight open-source model optimized for speed and low-resource deployment
Claude 3	Anthropic	Safety-first LLM with strong performance in summarization, Q&A, and ethical reasoning
LLaMA 3	Meta	Open-source model with multilingual support

THANK YOU!!!